

Eczema (Dermatitis)

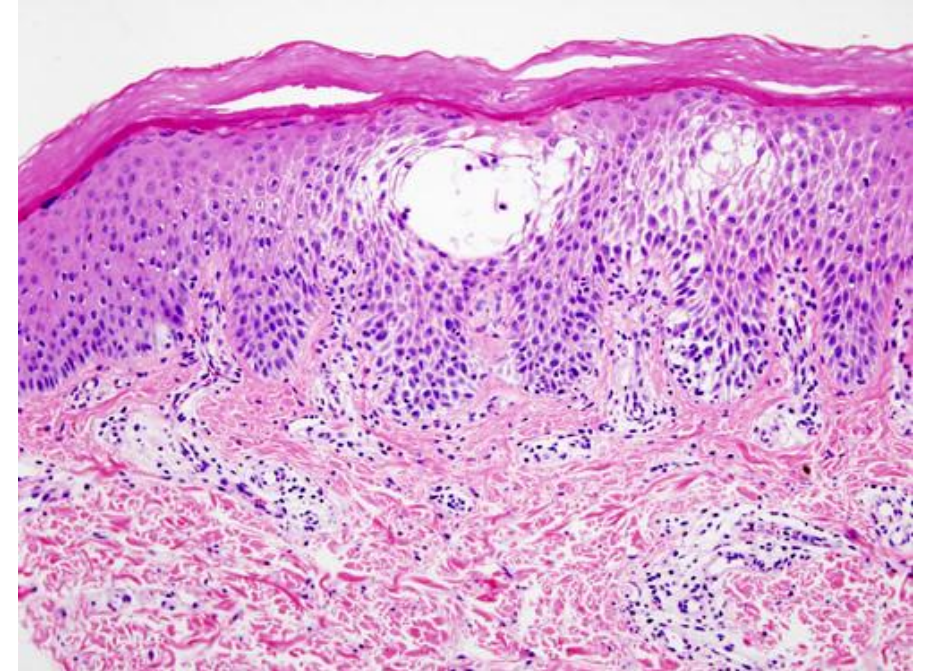
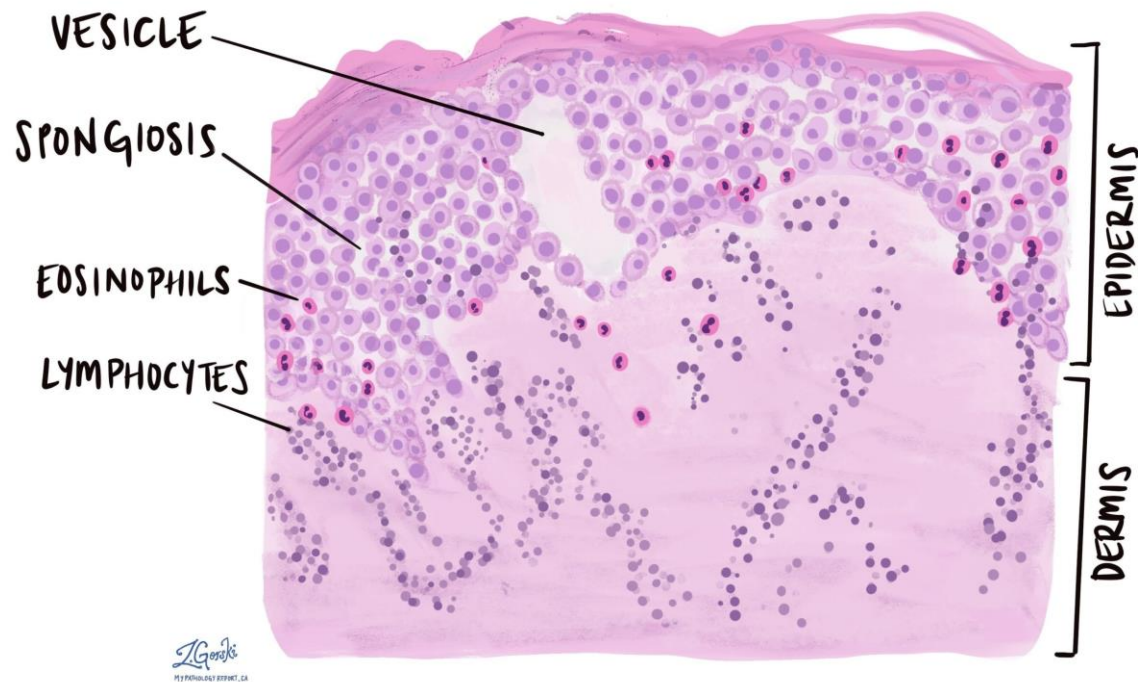
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Introduction:

- Eczema and dermatitis describes a pattern of inflammation in the skin
- Eczema and dermatitis are terms used to describe the characteristic clinical appearance of inflamed, dry, occasionally scaly, and vesicular skin rashes associated with multiple underlying causes
- Eczema is a clinical term but dermatitis is a pathological term
- Any eczema is dermatitis, but not any dermatitis is eczema.

Pathology:

- The clinical changes associated with dermatitis are reflected accurately at the cellular level.
- There is edema in the epidermis leading to **spongiosis** (separation of keratinocytes) and **vesicle formation**.
- The epidermis is **hyperkeratotic** (thickened) with dilated blood vessels and an **inflammatory (Lymphocytes & eosinophil) cell infiltrate** in the dermis.



Classification of eczema:

According to duration:

1. Acute eczema
2. Subacute eczema
3. Chronic eczema

According to the cause:

1. Endogenous eczema: An internal cause of the skin is responsible for the disease.
2. Exogenous eczema: An external cause is responsible for the disease
3. Mixed eczema: Both endogenous & exogenous processes precipitate the disease (e.g. Xerotic eczema)

According to duration:

1-Acute Eczema:

Characterized by:

- Intense pruritus
 - Intense erythema
 - Vesicular/bullous lesions and edema
 - Exudates (oozing).
- Secondary bacterial infections (staphylococcus and streptococcus) heralded by **golden crusting** may exacerbate acute eczema.



Clinical features:

2- Subacute Eczema:

- Erythema (lesser than in the acute stage).
- Crusting and scaling
- Fissuring
- Slight to moderate itching



Clinical features:

2- Chronic Eczema:

Leads to:

- Increased scaling,
- Increased xerosis (dryness)
- Lichenification (hyperpigmentation and thickening of the skin where surface markings become more prominent) i.e. the skin looks rather like the bark of a tree.



Eczema secondary changes:

Eczema is characteristically itchy and subsequent scratching may also modify the clinical appearance leading to:

- Excoriation marks
- Loss of skin surface
- Secondary infection
- Exudates
- Lichenification.
- Inflammation in the skin can result in disruption of skin pigmentation causing post-inflammatory hyper/hypopigmentation. (Patients often fear that loss of pigment is due to the application of topical steroids but in the majority of cases it is due to chronic inflammation).



According to the cause:

1. Endogenous (constitutional) eczemas:

- Atopic dermatitis (**Most common**)
- Seborrheic dermatitis
- Nummular (discoid) eczema
- Stasis (varicose) eczema
- Asteatotic eczema
- Dyshidrotic eczema (pompholyx)
- Neurodermatitis
- lichen simplex chronicus

2. Exogenous (contact) eczemas:

- Irritant contact dermatitis.
- Allergic contact dermatitis.
- Photodermatitis.

Endogenous eczemas:

Atopic Dermatitis:

- A chronic, pruritic, inflammatory skin disease that occurs most frequently in
- Associated with filaggrin gene defects which lead to impaired skin barrier function. This impaired barrier leads to increased transepidermal water loss and increased risk of antigens and infective organisms entering the skin.)
- Imbalance between T helper 2 (Two high in aTopic) vs Th1 and Th 17 (high in psoriasis).



Atopic Dermatitis:

- Often associated with:
 - An elevated serum **Immunoglobulin E (IgE)** level.
 - Personal or family history of **atopy**; A genetically mediated predisposition to an excessive (IgE) reaction encompassing a triad of **Eczema, Asthma & Allergic rhinitis**.
- **Atopy is type 1 hypersensitivity reaction.**



Atopic Dermatitis:

- Typically presents in infancy or early childhood, initially with facial and subsequently flexural limb involvement.
 - Babies and young children >> Face (esp. cheeks) and extensors
 - Older children and adults >> Flexures
- AD is intensely itchy, and even young babies become highly proficient at scratching, which can lead to disrupted sleep (both patient and family), poor feeding, and irritability.
- In severely affected babies **failure to thrive** may result.

Distribution of AD by Age

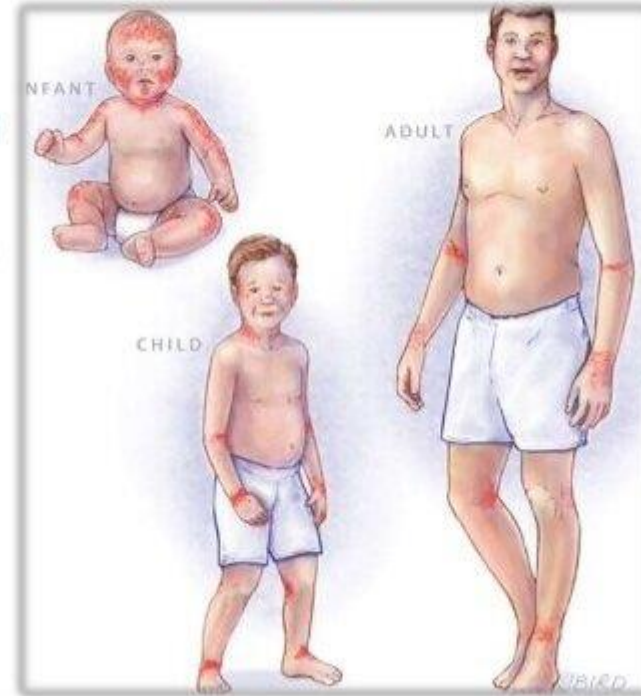
Infant

(birth-2 years)

Face (cheeks),
scalp, ears

Extensor
extremities

Seborrheic
dermatitis
overlap



Childhood

(2 years-puberty)

Face (cheeks)
Flexural extremities

Teenager-Adult

Localized flexural
extremities

Hands, dorsum feet

Birth – 2 years of life



Older than 2 years old



Atopic Dermatitis:

- The usual pattern is one of **flare-ups** followed by **remissions**, exacerbations being associated with:
 - 1- Inter-current infections
 - 2- Teething
 - 3- Food allergies.
 - 4- Climate (extremes of temperatures) and increased sweating
 - 5- Skin irritation
 - 6- Stress
 - 7- House dust mites

Atopic Dermatitis:

- In **older children or adults**, AD may become chronic and widespread and is frequently exacerbated by stress.
- AD is common, affecting 3% of infants; nonetheless, **90% of the patients spontaneously remit by puberty.**
- Patients likely to suffer from chronic AD in adult life are those who:
 - 1- Have a strong family history of eczema.
 - 2- Present at a very young age with extensive disease.
 - 3- Have associated asthma/multiple food allergies.

Atopic Dermatitis associated features:



White dermographism



Dennie-Morgan infraorbital folds



Palmar hyperlinearity



Centrofacial pallor

Atopic Dermatitis associated features:



Juvenile plantar dermatosis:
A variant of atopic eczema in which there is dry cracked skin on the forefoot in children .



Keratosis pilaris



- Thinning or absence of the lateral portion of the eyebrows (Hertoghe's sign)
- Periorbital darkening

Atopic Dermatitis associated features:



Pityriasis alba: variant of atopic eczema in which pale patches of hypopigmentation develop on the face of children



Nipple eczema

Atopic dermatitis complications:

- Secondary bacterial infection causing impetigo (m.c)
- Viral warts
- Eczema herpeticum is a life-threatening complication

Eczema herpeticum:

- It is a **herpes simplex** viral infection superimposed onto the skin affected by eczema (usually in atopic dermatitis patients).
- There is frequently a history of close contact with an adult with herpes labialis (cold sore).
- Clinically, there are multiple small ‘**punched-out**’ looking **ulcers**, especially around the neck and eyes.
- Eczema herpeticum is a serious complication of eczema that may be **life-threatening**, and therefore early intervention with **systemic aciclovir** is essential.



Emergency.



Atopic dermatitis treatment:

- Identify and avoid irritants
- Emollients at bath time
- Topical steroids, mild for young children, potent for older
- If 2ry infection occurs then use systemic antibiotics
- Antihistamines, Tacrolimus (topically), cyclosporine, and UV light may be used.

Seborrheic dermatitis:

- Chronic eczema characteristically involves areas rich in sebaceous glands with high sebum production (scalp, face, and upper trunk) and large body folds e.g. groin.
- Chronic, relapsing, and usually mild form of dermatitis that occurs in infants and in adults.
- The severity may vary from minimal, asymptomatic scaliness of the scalp (**dandruff**) to more widespread involvement.
- Biphase incidence with a **male predominance** in adults:
 - 1- Infants between the ages of 2 weeks and 12 months
 - 2- Adolescence and adulthood
- Occasionally associated with **HIV infections, Parkinson's disease & other neurological diseases**.

Seborrheic dermatitis:

Pathogenesis:

- Not entirely known
- Occurs in body sites with an increased number of sebaceous glands.
- May involve the lipid-dependent fungus *Malassezia furfur* which thrives on sebum thus causing dermatitis.

Seborrheic dermatitis:

Clinical manifestations In infants:

- Scalp (vertex and frontal areas; the **‘cradle-cap’** area)
- Face (forehead, eyebrows, eyelids, nasolabial folds, temple)
- Diaper area
- Retroauricular folds
- Neck
- Axillae



Seborrheic dermatitis:

Clinical manifestations in adults:

- **Scalp:** The earliest sign is **dandruff**, accompanied later by itching & inflammation, and retro-auricular fissuring.
- **Face:** Scaling & erythema of forehead, medial portion of eyebrows, eyelids, nasolabial folds, lateral part of the nose, and retro-auricular region.
- **Trunk:** Papules, greasy scales.
- **Flexural areas:** Erythema, greasy scaling, and secondary infection.



Seborrheic dermatitis:

Management:

- Treatment over many years with no definitive cure.
- Topical hydrocortisone is effective >> Recurrence after discontinuation.
- Steroid lotions/tar shampoos for the scalp.
- Ketoconazole shampoo/cream.
- Imidazole/hydrocortisone combinations.
- Maintenance is needed.

Asteatotic eczema:

- Occurs in **older** people with dry skin, particularly on the **lower legs**.
- The pattern on the skin resembles a **dry riverbed** or **'crazy-paving'**.
- RF: Low humidity: desert, high altitude, Excessive bathing especially using soaps and detergents



Discoid (nummular) eczema:

- Appears as intensely pruritic **coin-shaped** lesions.
- Most commonly on the **limbs**.
- Lesions may be vesicular
- Lesions frequently colonized by *Staphylococcus aureus*.
- **Males** are more frequently affected than females.
- Treatment: Emollient and potent corticosteroid ointment



Dyshidrotic eczema (pompholyx):

- Itching deep-seated vesicles on the **fingers, palms, and soles**.
- The blisters are small, firm, intensely itchy, and occasionally painful.
- The condition is more common in patients with nickel allergy.
- Cheiropompholyx: endogenous eczema of palms
- Pedopompholyx: endogenous eczema of soles



Venous (stasis) eczema:

- It is a common insidious dermatitis that occurs on the lower legs of patients with venous insufficiency.
- These patients have backflow of blood from the deep to the superficial veins leading to venous hypertension.
- In the early stages, there is brown haemosiderin pigmentation of the skin, especially on the medial ankle, but as the disease progresses skin changes can extend up to the knee.
- Patients typically have peripheral edema, and ulceration may result.
- **The mainstay of management is compression** + treatment of underlying cause



Investigations of eczema:

1- **Skin swabs** should be taken from the skin if a secondary bacterial or viral infection is suspected.

- The swab should be moistened in the transport medium before being rolled thoroughly on the affected skin, coating all sides of the swab to ensure an adequate sample is sent to the laboratory.
- A significant growth of bacteria reported with its sensitivity and resistance pattern can be useful in guiding antibiotic usage.

2- **Nasal swabs** should be performed in older children and adults with persistent facial eczema to check for nasal *Staphylococcus* carriage (MRSA).

3- If a secondary fungal infection is suspected then **scrapings or brushings** can be taken for mycological analysis.



Investigations of eczema:

4- **Routine blood tests** are not necessary; however, an eosinophilia and raised immunoglobulin E (IgE) level may be seen.

5- **RAST** (radioallergosorbent testing) looks for specific IgE levels against suspected allergens such as aeroallergens (pollens, house dust mite and animal dander) and foods (egg, cow's milk, wheat, fish, nuts and soya proteins).

6- **Skin prick testing** may also be used to determine any specific allergies to aeroallergens or foods.

Investigations of eczema:

7- **A skin biopsy** (usually a punch biopsy) for histological analysis may be performed if the diagnosis is uncertain.

(e.g. unilateral eczema of the areola, which could be Paget's disease of the nipple).

8- Varicose eczema (leg ulcer) patients should have their **ABPI** (ankle brachial pressure index) measured before compressing their legs with bandages. ABPI is the ratio of their arm: ankle systolic blood pressure.



Exogenous eczema

Contact dermatitis:

- Contact dermatitis can result from irritant or allergic reactions in the skin.
- Patients with abnormal skin barrier function (e.g. those with eczema) are more likely to develop contact dermatitis and suffer from irritant reactions than those with normal skin.
- Clinical appearance of both allergic and irritant contact dermatitis may be similar, but there are specific changes that help in differentiating them.
 - **Allergic dermatitis:** tends to be intensely itchy and results in erythema, edema, and vesicles. The more chronic lesions are often lichenified.
 - **Irritant dermatitis (m.c)** may be itchy or sore and presents as slight scaling, erythema, and fissuring. Sharply demarcated lesions,

Irritant contact dermatitis:

- A primary irritant is a substance which, if applied in high enough concentration to normal skin is capable of producing an eczematous response following a **single exposure**.
- A primary irritant will cause eczema in everyone if it is applied in sufficient concentration for a sufficient time.
- This may be chemical or physical.
- Physical irritants include air conditioning, prosthetic limbs, personnel protective clothing, and repetitive mechanical trauma.
- Chemical irritants include detergents, solvents, and acids.
- Dermatitis occurs **soon after exposure** and the severity varies with the quantity, concentration, and length of exposure to the substance concerned.
- **Previous contact is not required**, unlike allergic dermatitis where previous sensitization is necessary

Irritant contact dermatitis:

Examples:

- ❑ **Wear and tear eczema** in housework and in many industries
- ❑ **Napkin dermatitis** (nappy rash) is another type of this eczema. In this type of eczema, the **skin folds are spared**, whereas the reverse is true for candidosis. This is due to prolonged contact with urine or feces, and results of ammonia production by bacteria.



Allergic contact dermatitis:

- This is due to the development of **delayed hypersensitivity (type 4 allergy)** to a specific chemical (sensitizer or allergen).
- Such allergens will **not** cause eczema even in high concentration in a normal person, but severe eczema may be provoked by brief exposure to a very low concentration in a **sensitized person**.

Allergic contact dermatitis:

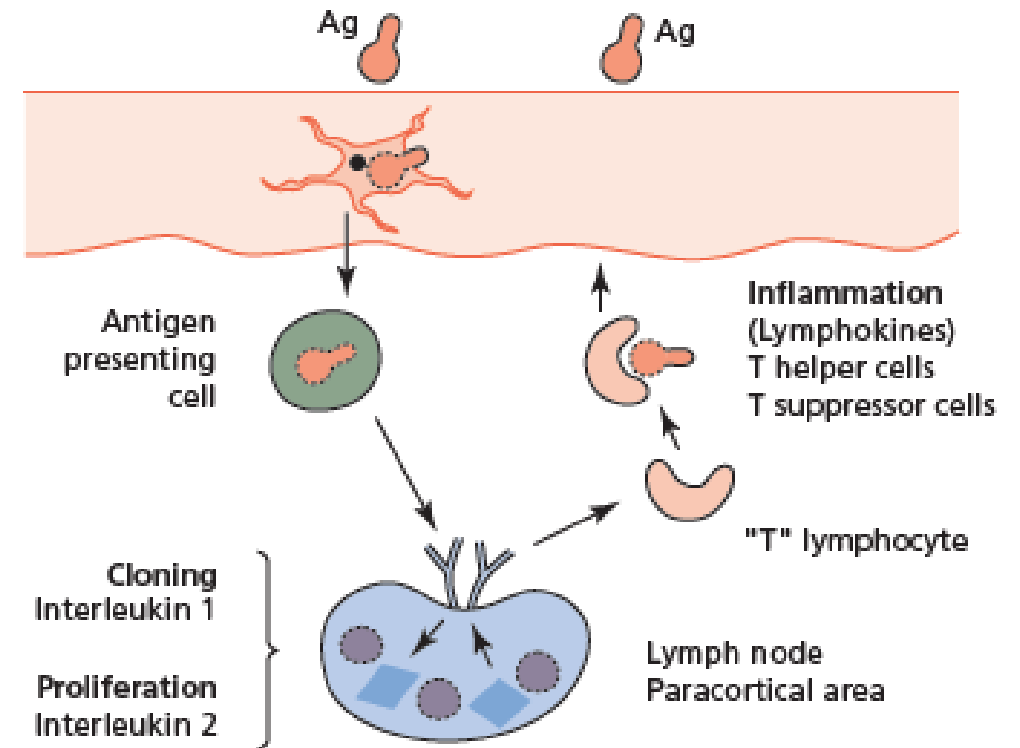
The characteristics of allergic dermatitis are as follows:

- Previous exposure to the substance concerned.
- 48–96 h between contact and the development of changes in the skin.
- Activation of previously sensitized sites by contact with an allergen at a distant skin site.
- Persistence of the allergy for many years.

Allergic contact dermatitis:

Immune mechanisms

- Allergic dermatitis results from **a type IV delayed hypersensitivity reaction** in the skin.
- Specific antigens (usually proteins) penetrate the epidermis, combine with a protein mediator, and are then picked up by Langerhans cells. This causes T-lymphocytes in regional lymph nodes to become sensitized to the antigen. On subsequent exposure to the antigen, an allergic reaction occurs because of the accumulation of sensitized T-lymphocytes at the site of the antigen with a resultant inflammatory response.
- This takes **48 h** and is amplified by interleukins that provide a positive feedback stimulus to the production of further sensitized T-lymphocytes



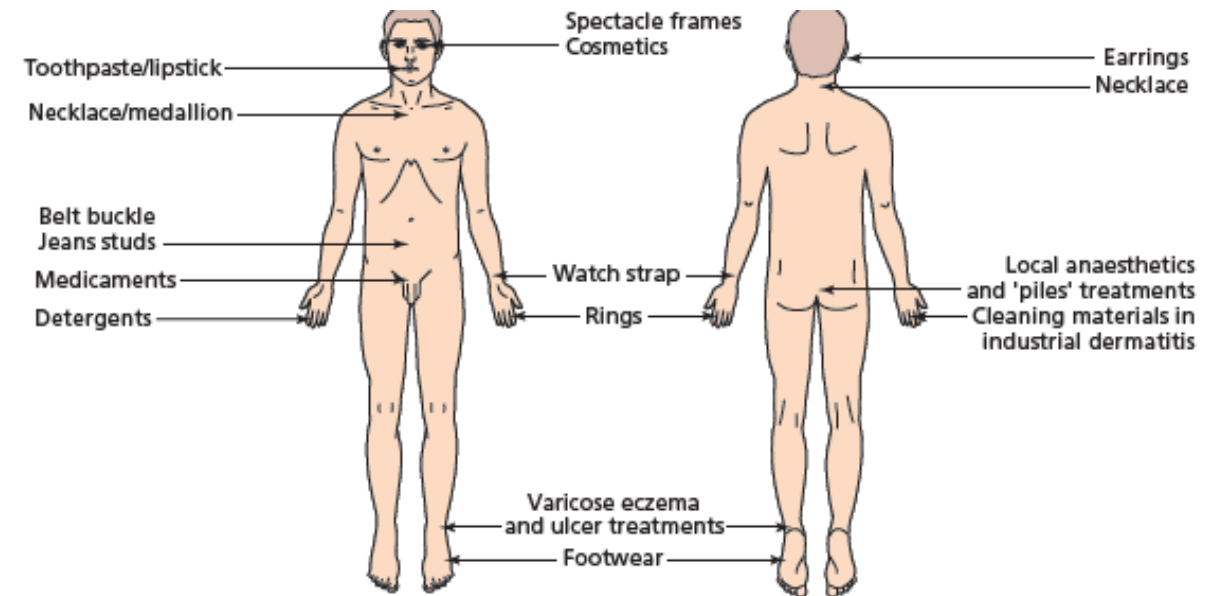
Contact dermatitis:

Box 4.1 Common contact allergens

- Nickel/cobalt (jewellery, clothing, wristwatch, scissors and cooking utensils).
- Potassium dichromate (chemical used to tan leather; Figure 4.17).
- Perfumes, myroxylon pereirae (balsam of Peru, fragrances; Figure 4.20).
- Formaldehyde, parabens, quaternium, methylchloroisothiazolinone, methylisothiazolinone (MCI/MI) (preservatives).
- PPD (permanent hair dyes, temporary tattoos and textiles; Figure 4.22).
- Ethylenediamine (adhesives and medications).
- Chromates (cement and leather).
- Mercaptobenzothiazole, thiurams (rubber gloves and shoes).
- Neomycin, benzocaine (medicated ointments; Figure 4.21).
- Lanolin (wool alcohol, emollients and mediated ointments).

Contact dermatitis clinical features:

- The distribution of the skin changes is often helpful in identifying the underlying cause.



Contact dermatitis clinical features:

- m.c allergen: **Nickel** (e.g watches and Jewlery)
- For example, an itchy rash on the foot may indicate an allergy to **footwear** such as an allergy to **potassium dichromate**, a chemical used to tan leather.
- An allergy to medications used for treating leg ulcers is a common cause of persistent dermatitis on the lower leg, or **iodine** used to clean the skin pre-operatively.



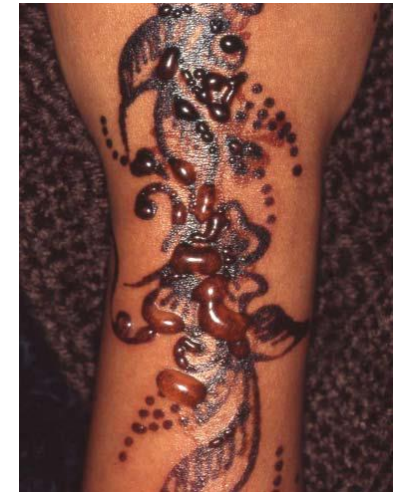
Contact dermatitis clinical features:

- Hand dermatitis can result from:
 - Glove **allergies** (rubber and latex)
 - Contact with an occupational exposure such as melamine formaldehyde resin **allergen**
 - **Irritation** from sweat under gloves.
- Fragrance allergy is common and can cause reactions to cosmetic products.



Contact dermatitis clinical features:

- Patients can develop allergies to medications such as those containing neomycin.
- Allergy to paraphenylenediamine (PPD) in a tattoo causes a very brisk vesicular reaction.
- Patients may also develop contact allergies to dressings/plasters applied to the skin.



Differences Between Irritant and Allergic Contact Dermatitis*

		Irritant CD	Allergic CD
Symptoms	Acute	Stinging, smarting → itching	Itching → pain
	Chronic	Itching/pain	Itching/pain
Lesions	Acute	Erythema → vesicles → erosions → crusts → scaling	Erythema → papules → vesicles → erosions → crust → scaling
	Chronic	Papules, plaques, fissures, scaling, crusts	Papules, plaques, scaling, crusts
Margination and site	Acute	Sharp, strictly confined to site of exposure	Sharp, confined to site of exposure but spreading in the periphery; usually tiny papules; may become generalized
	Chronic	Ill-defined	Ill-defined, spreads
Evolution	Acute	Rapid (few hours after exposure)	Not so rapid (12–72 h after exposure)
	Chronic	Months to years of repeated exposure	Months or longer; exacerbation after every reexposure
Causative agents		Dependent on concentration of agent and state of skin barrier; occurs only above threshold level	Relatively independent of amount applied, usually very low concentrations sufficient but depends on degree of sensitization
Incidence		May occur in practically everyone	Occurs only in the sensitized

*Differences are printed in bold.

Photodermatitis:

- Photodermatitis is caused by the interaction of light and chemicals absorbed by the skin.
- It can result from:
 1. Systemic medications, such as sulphonamides, phenothiazines, tetracycline, and voriconazole
 2. Substances in contact with the skin, such as topical antihistamines, local anesthetics, cosmetics, and antibacterials.



Phytophotodermatitis:

- Is due to contact with plant material, often containing forms of psoralens (poison oak, common rue, lime juice, and celery) and sunlight causing allergic contact dermatitis.



Occupational dermatitis:

1. The dermatitis first occurred during specific employment and had not been present before.
2. The condition generally improves or clears when away from the workplace.
3. There is exposure to known irritant/allergic substances and personnel protective measures are inadequate.

Contact urticaria:

- Is an immediate-type sensitivity reaction that can occur to certain food proteins and latex glove allergy.



Patch testing:

- Patch testing is used to determine the substances that cause allergic contact dermatitis.
- The concentration used is critical to ensure a low false negative/positive rate.
- The optimum concentration and best vehicle have been ascertained for the most common allergens.
- The standard series contains a group of tests that encompasses the **most common allergens encountered**.
- Done by dermatologists in outpatient settings.



Patch testing:

- The test patches are usually placed on the upper back (sites marked) and left in place for **48 h**, then removed and any positive reactions are noted.
- A further examination is carried out at **96 h** to detect any late reactions.
- Patients need to visit the unit **three times in 1 week**, be off systemic immunosuppressants and have an area of clear skin (usually the upper back) on which to perform the tests.



Patches being removed after 48 h.



Positive patch test reactions

General management of eczema:

- We must always remember to treat the patient and not just the rash.
- Exogenous eczema may be transient and ‘cured’ after **identification and avoidance of trigger factors.**
- Treatments usually start with simple emollients and mild topical steroids but with increasingly recalcitrant disease stronger therapies such as systemic immunosuppressants may be needed.

General management of eczema:

1. Emollients:

- In general, the oilier the cream the better its emollient properties; however, patients tend to prefer lighter creams.
- **Best is to use ointments.**
- Moisturizers should be applied repeatedly throughout the day in generous amounts.
- Emollients help to restore barrier function and reduce itching.

2. Cleansers:

- Normal soap contains surfactants that disrupt the lipid barrier and cause drying of the skin.
- **Aqueous cream** and **emulsifying ointments** are useful soap substitutes.
- Antibacterial moisturizing washes are very useful in those prone to infected eczema.
- Bath oils/shower washes that help prevent the drying of the skin caused by water alone.
- Soothing and de-scaling shampoos are useful for dry, flaky, and itchy scalps.

3. Topical steroids – frequency and quantity:

- Mainstay of treatment for active eczema.
- Ointments rather than creams should be used whenever possible (creams are more likely to cause irritation and have a higher risk of inducing contact allergies).
- Steroids should be applied once or twice daily to the affected skin only.
- One finger-tip unit (a line of ointment from the tip of the finger to the first skin crease) is a sufficient amount to treat a hand-sized (palmar and dorsal surface) area of affected skin.

Topical steroids – potency:

- **For Mild Eczema:**

- >> **Very low-potency** steroids such as hydrocortisone can be used.

- **For moderate to severe disease:**

- >> **potent** topical steroids (mometasone, betamethasone, and fluocinolone acetonide) for short periods >> followed by **steroid ‘holidays’** (Better than using daily low potency steroids)

- **For acute dermatitis:**

- >> Start with a **potent** topical steroid for a few days or weeks and then >> reduce to a **lower** potency once the disease is controlled.

- **For Face and groin:**

- >> **Lower** potency topical steroids should be used.

4. Immunomodulators:

- Useful topical preparations to control mild eczema.
- Types:
 - 1- Tacrolimus (0.03% – children aged 2–15 years, 0.1% – adults)
 - 2- Pimecrolimus (1%)
- Should be applied twice daily to the affected skin.
- Their use should be limited to those who have **failed to respond adequately to first-line treatment** (topical steroids).
- It is recommended that their use be for a **short period** rather than continuous use.
- There is a theoretical risk of **increased malignancies**.

5. Occlusion:

- Covering topical therapy with bandages, body suits, ‘wet-wraps’ and dressings can be very helpful in the management of chronic eczema.
- Prior to occlusion, the practitioner should ensure the eczema is **not infected**.
- Generally **worn overnight**.
- Occlusive therapy helps **relieve symptoms of itch, keep emollient creams on the skin, and ‘drive’ topical therapy through the epidermis.**
- The potency of topical steroids is **enhanced 100-fold by occlusion**; therefore, only very low-potency steroids should be used under occlusion.

6. Antibiotics:

- Occasionally, antibiotics are needed to treat **infected eczema**; they may be given topically or systemically.
- **Topical antibiotics:**
 - e.g., **fusidic acid**, silver sulfadiazine, polymyxins, neomycin and **mupirocin**.
 - It is recommended that topical antibiotics should be used for a **maximum of 2 weeks** continuously to try to reduce the risk of developing resistant bacteria.
- **Systemic antibiotics:**
 - e.g., flucloxacillin (amoxicillin and penicillin) erythromycin (clarithromycin and azithromycin), and ciprofloxacin (levofloxacin and ofloxacin).
- Antibacterial emollient washes can be useful in managing active cutaneous infections as well as preventing them by daily use.

7. Phototherapy:

- Narrow-band UVB or PUVA (psoralen with UVA) can be highly effective for generalized eczema.
- Each course of phototherapy lasts 6–8 weeks with patients attending 2–3 times per week.

8. Systemic therapy:

- Severe widespread disease not controlled with topical therapy may require systemic immunosuppressants.
- Rapidly reducing course of oral prednisolone (30 mg for 5 days, then reduced by 5mg every 5 days) to control very severe generalized acute eczema.
- Long-term oral steroids should not be used to control eczema.
- Rather, azathioprine, ciclosporin, mycophenolate mofetil, and methotrexate can be used for long-term management.
- If available, levels of thiopurine methyl transferase (TPMT) should be checked prior to initiation of azathioprine.

Thank You